

Dr. Nezah Balal

Senior Lecturer, Department of Electrical and Electronics Engineering
 Ariel University
 Applied Magnetic Systems Laboratory
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Academic Profile

Dr. Nezah Balal is a Senior Lecturer in the Department of Electrical and Electronics Engineering at Ariel University and Head of the Applied Magnetic Systems Laboratory. His research focuses on millimeter-wave and micro-Doppler radar, terahertz radiation sources, periodic magnetic systems and undulators, and high-frequency antennas and RF components, including 3D-printed devices. His work combines theory, simulation, fabrication, measurements, and signal processing, with applications in remote sensing, target classification, speech and vital-sign monitoring, and compact electromagnetic systems.

Research Interests

- Millimeter-wave and micro-Doppler radar systems
- Contactless human sensing and radar target classification with machine learning
- Terahertz radiation sources and free-electron devices
- Periodic magnetic systems and helical undulators
- High-frequency antennas, RF circuits, and 3D-printed RF components

Academic Appointments

2024–Present	Coordinator of B.Sc. Final Projects (Capstone), Department of Electrical and Electronics Engineering, Ariel University.
2022–Present	Senior Lecturer, Department of Electrical and Electronics Engineering, Ariel University.
2022–Present	Head, Applied Magnetic Systems Laboratory, Ariel University.
2021–2022	Senior Lecturer, ORT Braude College of Engineering.
2020–2021	Adjunct Lecturer, Ariel University and Technion – Israel Institute of Technology.
2018–Present	Adjunct Senior Teaching Fellow, Department of Physics, Technion – Israel Institute of Technology.
2017–2020	Postdoctoral Research Fellow (Levi Eshkol Postdoctoral Fellowship).

Education

2012–2016	Ph.D. in Electrical Engineering. Thesis: Efficient Mechanisms for Generation of Terahertz Waves in Free-Electron Devices and Their Applications. Supervisors: Prof. Vladimir Bratman and Prof. Yosef Pinhasi.
2010–2013	M.Sc. in Electrical Engineering. Thesis: Atmospheric and Fog Effects on Ultra-Wide Band Radar Operating at Extremely High Frequencies.
2006–2010	B.Sc. in Electrical Engineering.
2006–2009	B.Sc. in Applied Physics.

Academic and Professional Activities

2025	Guest Editor, Special Issue in <i>Electronics</i> (MDPI): “Terahertz Antenna Technologies for Next-Generation Wireless Systems”.
2025–Present	IEEE Senior Member. Patent applications and registrations in Israel and abroad in the field of magnetic systems.

Teaching and Supervision

Selected courses include Introduction to Radar Systems, Linear Systems, Microelectronic Circuits, and Introduction to Electrical Engineering. Supervises postdoctoral researchers, including Dr. Ayushman Ramola, and

collaborates with Prof. Michael D. Tokman in the Applied Magnetic Systems Laboratory. Also co-supervises Ph.D. and M.Sc. students in radar systems, magnetic systems, and terahertz technologies, and has supervised over 30 undergraduate final projects since 2011.

Selected Honors and Research Support

2024–2027	Principal Investigator, Ministry of Innovation, Science and Technology, Israel: Microscopic Magnetic Coils.
2023–2025	Co-Principal Investigator, Ministry of Defense, Israel: Muon Accelerator.
2018	Outstanding Teaching Assistant Award (faculty level), Faculty of Physics, Technion.
2017–2020	Levi Eshkol Postdoctoral Fellowship, Israeli Ministry of Science, Technology and Space.

Selected Publications

1. N. Steinmetz and N. Balal, “Feasibility Study of Real-Time Speech Detection and Characterization Using Millimeter-Wave Micro-Doppler Radar,” *Remote Sensing*, 17(1), 91, 2025. DOI: [10.3390/rs17010091](https://doi.org/10.3390/rs17010091).
2. Y. Richter and N. Balal, “High-Resolution Millimeter-Wave Radar for Real-Time Detection and Characterization of High-Speed Objects with Rapid Acceleration Capabilities,” *Electronics*, 13(10), 1961, 2024. DOI: [10.3390/electronics13101961](https://doi.org/10.3390/electronics13101961).
3. Y. Richter, N. Balal, and Y. Pinhasi, “Neural-Network-Based Target Classification and Range Detection by CW MMW Radar,” *Remote Sensing*, 15(18), 4553, 2023. DOI: [10.3390/rs15184553](https://doi.org/10.3390/rs15184553).
4. N. Balal, E. Bamberg, V. L. Bratman, and E. Magory, “Fabrication and Experimental Study of Prototype NdFeB Helical Undulators,” *IEEE Transactions on Electron Devices*, 70(11), 5911–5915, 2023. DOI: [10.1109/TED.2023.3315250](https://doi.org/10.1109/TED.2023.3315250).
5. N. Balal, I. V. Bandurkin, V. L. Bratman, E. Magory, and A. V. Savilov, “Negative-mass mitigation of Coulomb repulsion for terahertz undulator radiation of electron bunches,” *Applied Physics Letters*, 107(16), 163505, 2015. DOI: [10.1063/1.4934495](https://doi.org/10.1063/1.4934495).
6. A. K. Shakya, A. Ramola, N. Balal, and A. Bergman, “Evolution of Photonic Crystal Fiber-Based Smart Sensors with Emphasis on Biomedical Applications: A Review,” *IEEE Sensors Journal*, 26(4), 5288–5301, 2026. DOI: [10.1109/JSEN.2026.3650816](https://doi.org/10.1109/JSEN.2026.3650816).

Full publication list: [Google Scholar](https://scholar.google.com/citations?user=...).